DD 499 BDes Project IV

Product Photography Guide for Handloom Artisans

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Date: 30th April, 2025

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This is to certify that the Project Report entitled
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Abstract

This project explores the challenges faced by Indian handloom artisans in effectively presenting their products on digital platforms. Despite the rich cultural and material value of handloom textiles, artisans often struggle to visually communicate key fabric properties, such as texture, drape, and transparency, through product photography. Through a combination of desk research, user interviews, social media analysis, and observational studies, the project identifies major gaps in current visual representation practices. Building on these insights, the project proposes a solution in the form of a smartphone-based photography support system. The final design features guided templates, lighting instructions, real-time camera prompts, and video tutorials to help artisans capture high-quality product images using minimal resources. The solution aims to bridge the sensory gap between physical and virtual shopping experiences, ultimately improving artisans' digital presence and access to wider markets.

1. Introduction

1.1 Background and current scenario

India's artistic heritage stretches back millennia, with artisans forming the backbone of the country's cultural identity. Their skills, passed down through generations, are evident in everything from intricate textiles and pottery to majestic sculptures and dazzling jewellery.

Historically, artisans catered to the needs of royalty and wealthy patrons, creating exquisite pieces that reflected the opulence of the times. The Mughal era, for instance, saw a flourishing of craftsmanship with fine metalwork, carpets, and textiles becoming highly sought after.

However, in the modern world, artisans grapple with the fast pace of change. Rapidly evolving consumer trends and competition from mass-produced goods pose significant challenges. The 1995 census accounted for 6.5 million weavers and associated artisans. The numbers reduced by about a third to 4.3 million by the time the 2009 census rolled around. As per the last handloom census, 89% of handloom artisans currently earn less than Rs10,000 a month. In fact, as many as 67% earn less than Rs5,000. And yet, the next generation of artisans aspires to earn more than Rs10,000 in order to maintain a sustainable livelihood for their families, without this minimum income, they may not continue in this vocation.

Despite the unique value of handloom products, artisans often face significant challenges in accessing broader markets. Many still rely on traditional, offline methods to sell their products, with only a small fraction transitioning to online platforms. Even among those who go digital, most struggle to present their products in a visually impactful and appealing way.

According to the Handloom Census 2019–20, only 1% of weavers and artisans sell their products through online channels, and online sales account for just 0.2% of total handloom sales. Furthermore, despite the growing global demand for handcrafted products, less than 20% of handloom goods reach international markets.[1]

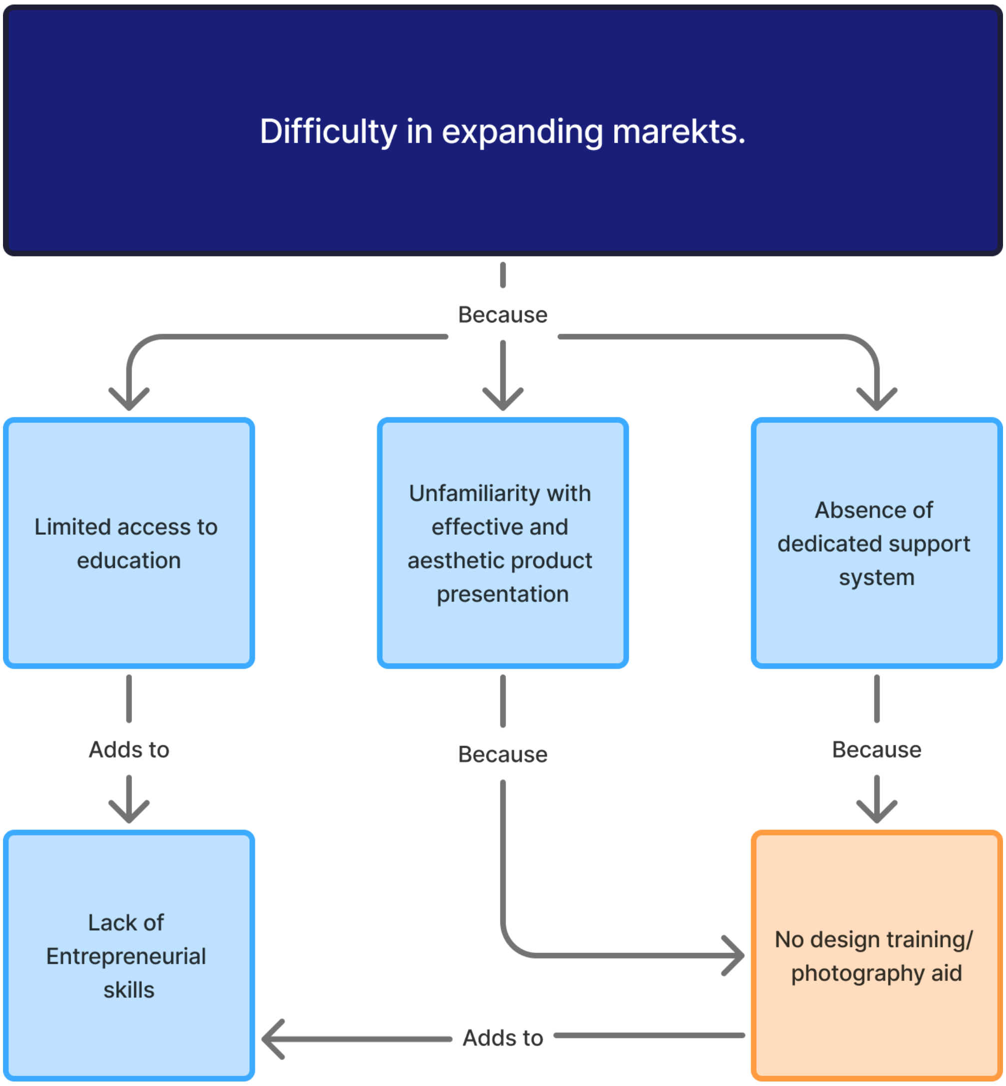

Image 1: Flow Diagram of Barriers Limiting Market Reach for Handloom Artisans

This gap in market reach can be attributed to several key factors.

- Firstly, a significant number of artisans have limited access to education (60.2% of male and 51.8% of female artisans in India have received only primary or no formal education) [1]. This lack of educational opportunity results in poor entrepreneurial and business skills, hindering their ability to sustain their craft as a viable source of income.
- Secondly, most artisans are unfamiliar with how to effectively and aesthetically present their products, especially in digital formats, which limits their appeal to modern buyers.
- Thirdly, the absence of dedicated support systems, such as design mentorship, digital marketing training, and access to selling platforms, prevents them from reaching broader markets, both within India and globally.

To better understand the purchasing patterns of the Indian population, especially in the context of handloom and apparel products, a detailed research study was conducted. The primary objectives of this research were twofold:

1. To identify the differences in user perception of fabric when experienced in virtual versus physical environments.

2. To determine the key information required to help consumers form a comprehensive understanding of fabric properties, enabling them to make confident and informed purchase decisions.

The research was carried out in two phases. It began with secondary research, which involved reviewing existing literature, analyzing market trends, and studying online fabric retail practices, user behavior in digital shopping contexts, and the role of sensory information in textile evaluation. As part of this phase, Instagram profiles of handloom artisans from various clusters in Assam were examined to identify common mistakes in product photography. Observations revealed recurring issues such as poor lighting, improper composition, suboptimal fabric presentation, and a lack of detailed information, all of which limited the effectiveness of their online presence.

Following this, primary research was conducted through semi-structured user interviews. Photographs of an apparel product were taken in various compositions and under different lighting conditions to simulate typical artisan photography scenarios. These images were then shown to participants, who were asked to evaluate the fabric based on key attributes such as texture, weight, softness, thickness, drape, and transparency. Subsequently, participants interacted with the physical product and re-evaluated the same attributes.

This comparative approach allowed us to draw meaningful insights into:

- **Which fabric attributes are most important to users** when assessing quality and suitability

- **How users perceive those attributes differently in virtual versus physical contexts**

- **What types of visual and descriptive cues help bridge this perception gap in online settings**

The findings highlighted the limitations of current digital fabric presentations and pointed to clear opportunities for enhancing the virtual representation of handloom products.

1.2 Aim of the project

The aim of the project is primarily to design a solution that addresses the challenges faced by Indian handloom artisans in showcasing their craft, with the goal of enhancing their market reach and the perceived value of their products in digital spaces.

1.3 Our Solution: Guiding Better Product Photography

The problem was addressed by developing support mechanisms to help artisans photograph their products more effectively. A set of product photography guidelines was created, informed by research on how users perceive fabric properties in digital media. These guidelines served as the foundation for the design of a user-friendly platform featuring photography templates and video tutorials intended to guide artisans during the image-capturing process.

The solution offers **three key benefits** for artisans:

1. **Improved Composition:** The platform provides visual cues and step-by-step guidance directly within the camera interface, helping artisans compose their photographs more professionally.

2. **Highlighting Key Fabric Attributes:** The templates are designed to help artisans emphasize the most important and relevant properties of their handloom fabrics such as texture, drape, transparency, and borders which are often missed in standard product photos.

3. **Professional Results with Simple Setups:** Artisans can use basic phone cameras and easily available household material**s** to create photographs that still look polished and professional, without the need for expensive equipment or studio setups.

2. Literature Review

2.1 Sales and distribution channels

The first part of the literature review focused on understanding the current sales and marketing channels used by Indian handloom artisans, as well as evaluating the benefits and limitations of these methods.[2] The review identified two major modes of marketing: **offline** and **online**.

Offline Marketing Channels

These traditional methods include:

- **Craft Fairs and Events**, which allow artisans to directly engage with customers.

- **Retail Stores and Boutiques**, offering artisans an extended reach by collaborating with retailers aligned with their aesthetic and target audience.

- **Pop-up Shops**, which serve as temporary physical retail spaces in high-footfall areas.

Benefits of offline channels include:

- Tangible product experience

- Immediate purchasing decisions

- Personal interaction and trust-building

Online Marketing Channels

With digital connectivity on the rise, several artisans are beginning to explore:

- **E-commerce websites**, offering artisans full control over branding and storytelling.

- **Online marketplaces** (e.g., Etsy, Amazon Handmade), which provide access to a large customer base.

- **Social media platforms** (e.g., Instagram, Facebook), which help artisans visually showcase their products and connect with broader audiences.

Benefits of online platforms include:

- Wider geographic reach
- 24/7 product availability
- Access to consumer insights and engagement metrics

A review of relevant literature by Faruque and Guha (2023) [3] provides further empirical insights into the marketing channel choices made by handloom micro-entrepreneurs in rural Assam. The study revealed that although retailer channels were found to be the most economically efficient, the majority of artisans (over 50%) still rely on travelling traders, which offer the least marketing efficiency. Factors such as low educational attainment, lack of access to credit, limited market information, and poor road connectivity were cited as key determinants influencing this choice.

Additionally, the study emphasizes the significant role of socio-economic and institutional factors in shaping artisans' marketing decisions. For instance, artisans with greater business experience, access to market information, and membership in weavers' groups were more likely to opt for direct-to-retailer or wholesale channels, which offer higher returns. However, artisans with limited resources often default to informal and intermediary-heavy channels, reducing their overall earnings.

These findings suggest that while online platforms hold promise, their effectiveness is contingent on the artisan's ability to present products in a clear and compelling manner. A major gap identified is the inability to effectively communicate key fabric properties through digital means, which restricts user trust and purchase confidence.

Considering the advantages of both offline and online methods, an optimal solution would integrate the benefits of both—leveraging the reach and accessibility of digital platforms while ensuring users receive detailed, high-quality information that mimics the physical shopping experience.

2.2 Digital Adoption and Artisan Resilience

The COVID-19 pandemic accelerated the adoption of digital technologies across various sectors, including the craft and handloom industries. A study featured in the *NIFT Journal of Fashion* (2022)[4] emphasizes the crucial role of online technology in enhancing resilience among handicraft artisans, particularly during the economic downturn brought on by the pandemic.

Digital platforms served as **critical enablers for survival**, allowing artisans to maintain visibility, connect with markets, and explore alternative sales channels. However, despite this shift, the transition was uneven due to several constraints, such as:

- Digital illiteracy
- Limited access to smartphones or stable internet
- Inadequate visual communication skills

Importantly, the study highlighted that product presentation, especially through digital photographs, was often poor, affecting the visibility and desirability of artisan-made goods. Most artisans lacked the knowledge to effectively showcase their products in a way that communicates quality, material properties, and craftsmanship.

The study also revealed a strong correlation between digital engagement and market access, noting that artisans who were able to adopt even basic digital tools saw improved outcomes in terms of sales and customer outreach. However, without support in areas like visual storytelling, photography, and content creation, the potential of digital platforms remained largely untapped.

These insights reinforce the need for solutions that not only provide digital access but also equip artisans with practical tools and guidance to visually communicate their craft.

2.3 Apparel Product Presentation in Digital Media

As consumer behavior continues to shift toward digital platforms, the visual and informational presentation of products has become a critical determinant of purchase decisions. Several studies have emphasized that the quality and completeness of product photographs and descriptions directly impact consumers' perception of value, confidence, and willingness to purchase online.

In the context of apparel, Özbek et al. [5] conducted a study examining how different types of product photographs and levels of information influenced purchasing behavior. The results showed that elements like fabric composition, waist and length measurements, fit description, and even visual cues such as detailed close-ups and mannequin photography played a vital role in increasing consumer trust and purchase intention. It was found that 67% of consumers agreed that fabric content affects their decision, and over 66% highlighted the importance of viewing products on live mannequins. However, the study also noted that most digital platforms failed to provide critical visual information, such as fabric texture, stretchability, and garment construction details, leaving a gap in digital consumer experience [5].

This gap becomes even more pronounced when applied to **handloom products**, where tactile and visual richness is a core part of value perception. Unlike mass-produced items, handloom textiles carry intricate detailing, unique weaves, and textures that are often poorly represented through conventional photographs.

2.4 Scope of the Project

Based on the literature review done so far, some major directions for the project were mapped in order to scope down appropriately for the timeline of this thesis project.

The scope of this project was defined to include the study of:

- Identifying challenges in product presentation, particularly in digital media, with emphasis on issues related to photography, lighting, and visual communication.
- Studying user perception of fabric properties when experienced digitally versus physically, and determining which attributes are essential for informed purchasing decisions.
- Exploring how digital technologies can aid in bridging the perception gap.
- Designing a photography support system.
- A solution in the digital space

3. Problem statement

How might we help handloom artisans showcase their craft to a wider audience in a visually compelling and informative way ?

4. Methodology

This project followed the **Double Diamond Design Process** developed by the UK Design Council, which divides the design journey into four key stages: *Discover, Define, Develop,* and *Deliver*. This model allowed for a structured yet flexible approach to identifying challenges.

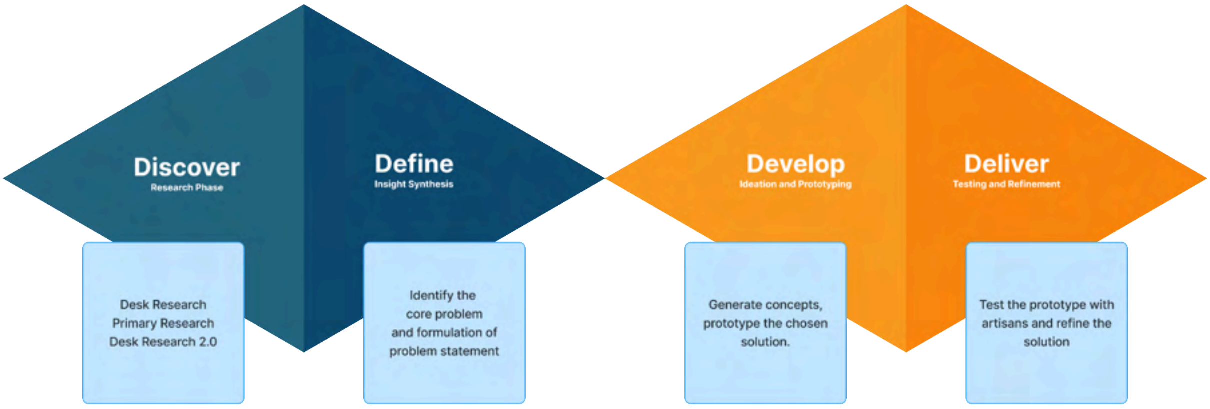

Image 2: Double Diamond Design Process for the project.

4.1 Discover (Research Phase)

The initial phase focused on building a deep understanding of the problem space.

- **Desk Research 1:** Existing literature, product presentation strategies of fashion brands and designers, and government data were reviewed to understand the economic status of handloom artisans, their existing sales channels (offline and online), and the challenges faced in accessing wider markets.

- **Primary Research:** Semi-structured interviews were conducted to study how users perceive fabric properties in digital images versus physical interaction. Photographs of apparel were taken under various lighting and composition setups, which were shown to participants for evaluation.

- **Desk Research 2:** Instagram profiles of handloom artisans, particularly from clusters in Assam, were analyzed to identify common issues in their product photography such as poor lighting, improper framing, and lack of clarity in highlighting fabric details.

4.2 Define (Insight Synthesis)

Insights from all research phases were synthesized to identify the core problem: **the gap in communicating key fabric properties through digital photographs**, create product photography guidelines and formulation of the problem statement that guided the rest of the project.

4.3 Develop (Ideation and Prototyping)

In this phase, ideas were generated to address the identified challenges. A solution was conceptualized in the form of a photography support system featuring visual templates and video tutorials. The system was prototyped with a focus on simplicity, accessibility, and relevance to the artisan's environment, keeping in mind the use of basic smartphones and local materials.

4.4 Deliver (Testing and Refinement)

The prototype was tested with users to evaluate its effectiveness in improving the quality and clarity of product photographs. Feedback was gathered on usability, visual impact, and whether key fabric attributes were better communicated. Based on this feedback, refinements were made to improve the templates and interface experience.

5. Phase 1: Discover

5.1 Desk Research

To explore how the product presentation of handloom artisans could be improved, it was essential to understand what established fashion brands and designers do differently, particularly in terms of visual communication. The aim was to identify how these brands create product photographs that convey a sense of quality and allow buyers to understand key product properties without experiencing the product in person.

As part of this investigation, a study was conducted on the Instagram profiles of 10 prominent fashion brands and designers, including Louis Vuitton, Gucci, Sabyasachi, Biba, iTokri etc. The analysis focused on how these brands showcase their products through photography, observing common visual strategies and drawing patterns.

Some of the observed similarities are:

1. Using Closeup Shots

It helps brands to highlight the texture, intricate embroidery, and fine details, offering followers a more visually immersive and tactile experience of the product.

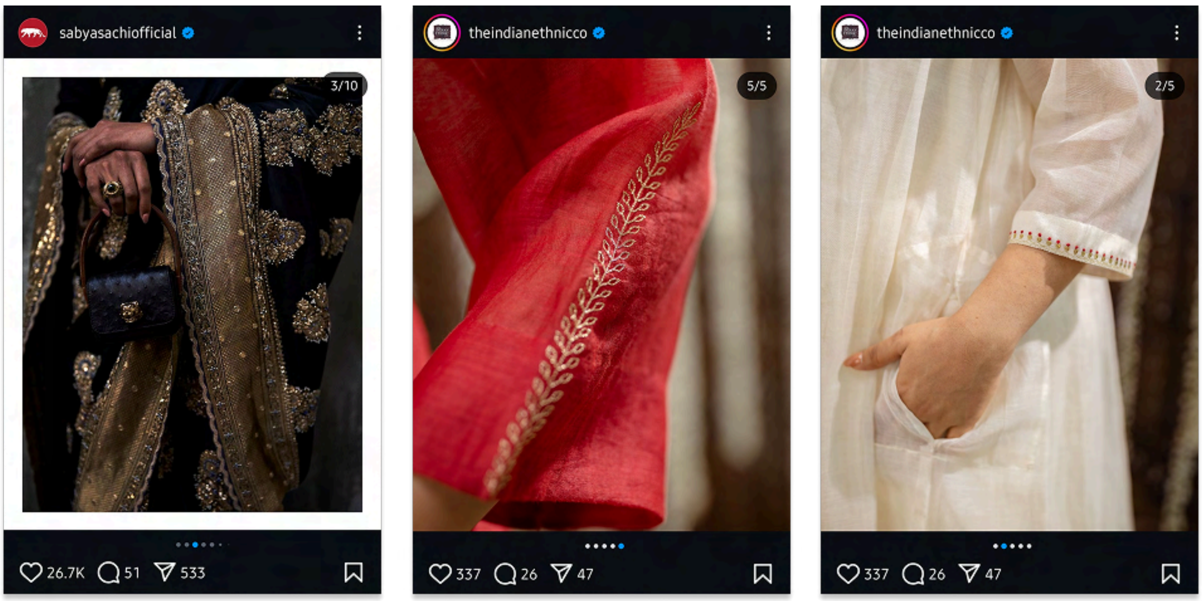

Image 3: Some Close-up shots of products posted by fashion brands and designers on Instagram

2. Shots from various angles

Brands showcase products from various angles on Instagram, giving users a comprehensive view and helping them better understand how the item will look, even from unusual perspectives.

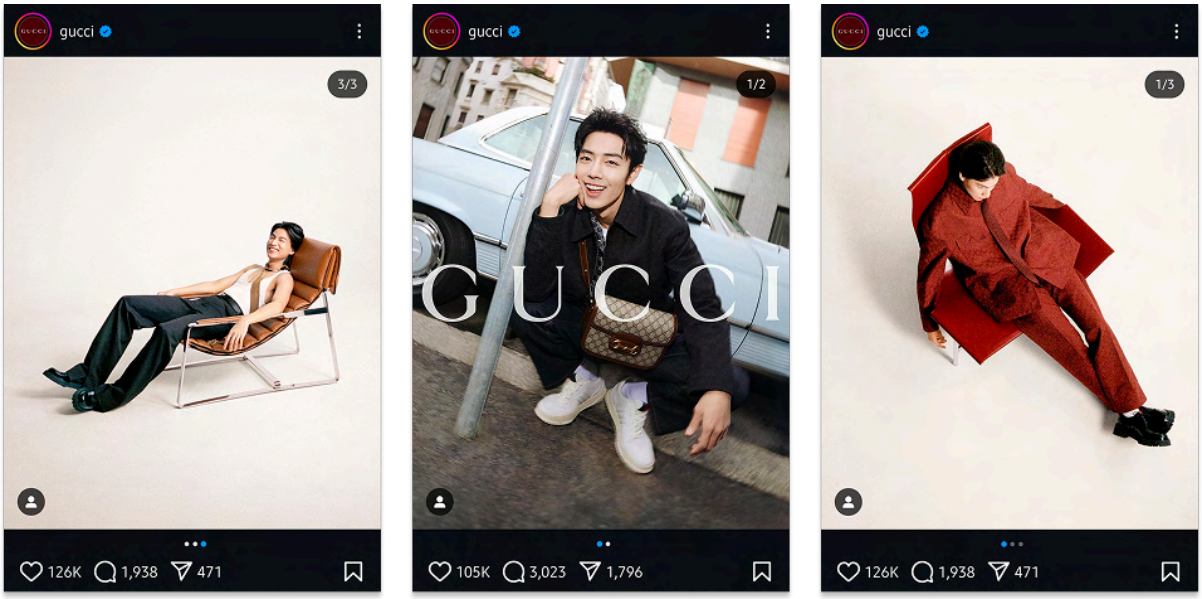

Image 4: Multiple angle shots of products posted by fashion brands and designers on Instagram

3. Diverse lighting

Brands display products under different lighting conditions (indoor, outdoor, daylight etc) to provide users with a realistic understanding of how the item will appear in various environments.

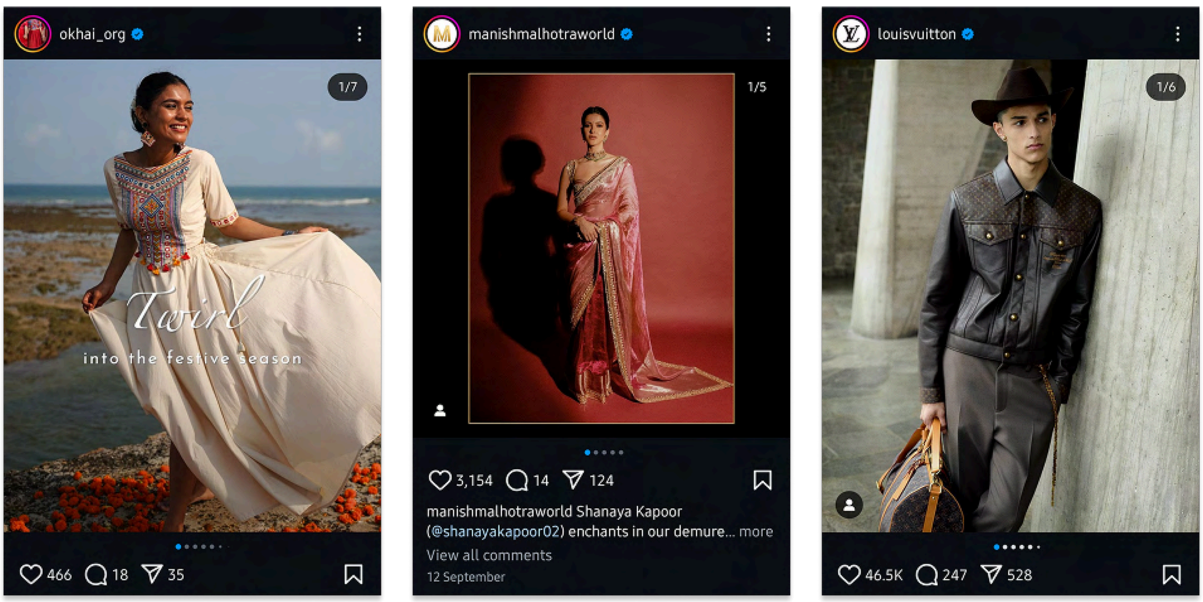

Image 5: Different lighting photos of products posted by fashion brands and designers on Instagram

4. Complementary backgrounds

Instead of using plain backgrounds, brands now utilize complementary backdrops that enhance the product's appeal and create a more visually engaging presentation.

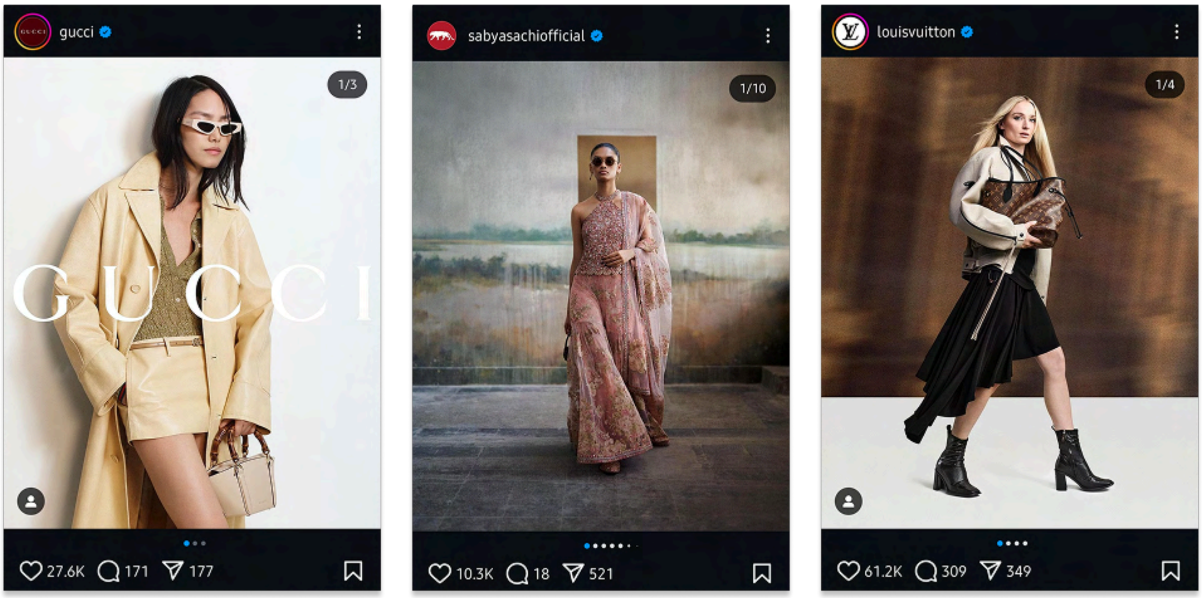

Image 6: Different backgrounds of products posted by fashion brands and designers on Instagram

5. Videos

Brands leverage product videos to showcase detail and storytelling elements to create a personal connection with viewers, engaging them emotionally and enhancing the overall narrative of the product.

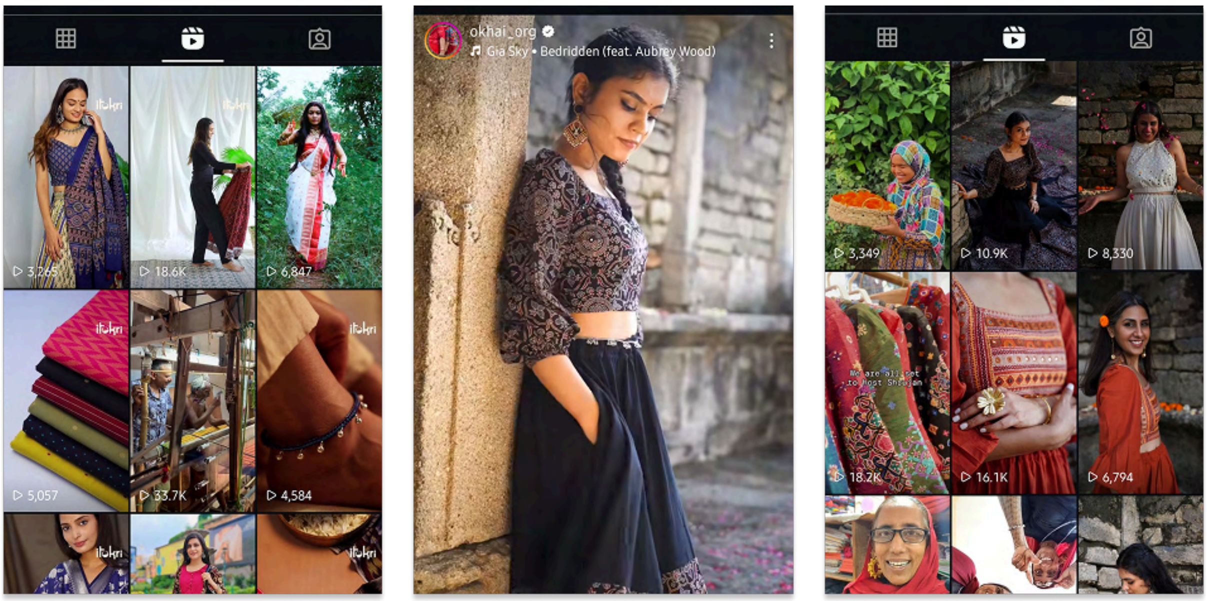

Image 7: Videos of products posted by fashion brands and designers on Instagram

The second part of the study focused on identifying the common mistakes made by handloom artisans while photographing their products. Instagram profiles of 10 handloom artisans from the Kamrup cluster were analyzed to observe recurring issues in their product presentation. This analysis helped in pinpointing specific areas where the visual communication of their products could be improved.

1. Artisans typically capture flat-lay pictures of their products by placing them on the floor or holding them up. While these images provide an overview of the design, they often fail to convey crucial material properties such as weight, texture, and transparency.

2. Artisans do not effectively showcase intricate motifs. Some attempt to, but make mistakes in composing pictures, such as incorrect fabric orientation or lack of emphasis on details.

3. Some artisans attempt to capture the flow and natural fall of the fabric to convey its weight, but poor composition often undermines their efforts. For products like stole, it is very difficult to get an idea of the size, because they lack juxtaposition.

4. Following a proper grid can improve the layout and alignment. Borders and edges of the fabric are often not highlighted or shown clearly, and when included, they are presented in a way that feels dull and unengaging.

5.2 Primary Research

To gain deeper insights into how users perceive fabric properties in digital versus physical contexts, semi-structured interviews were conducted with eight individuals who had experience shopping on online marketplaces. The study had two primary objectives:

1. To understand the differences between user perception of fabric when viewed virtually and physically.

2. To identify the key information users require to make informed purchasing decisions about fabric-based products.

The research approach involved photographing an apparel product in various lighting conditions and compositions. Each photograph was shown to the participants, who were then asked to evaluate the product based on properties such as texture, weight, drape, and transparency. Participants were also asked about the kind of visual information they would like to see before making a purchase decision online.

After this, participants were shown the actual product and were asked the same set of questions to compare their perception of the product in both virtual and physical settings. The responses were documented and analyzed to identify similarities, differences, and gaps in perception. These insights helped uncover opportunities for improving digital product representation.

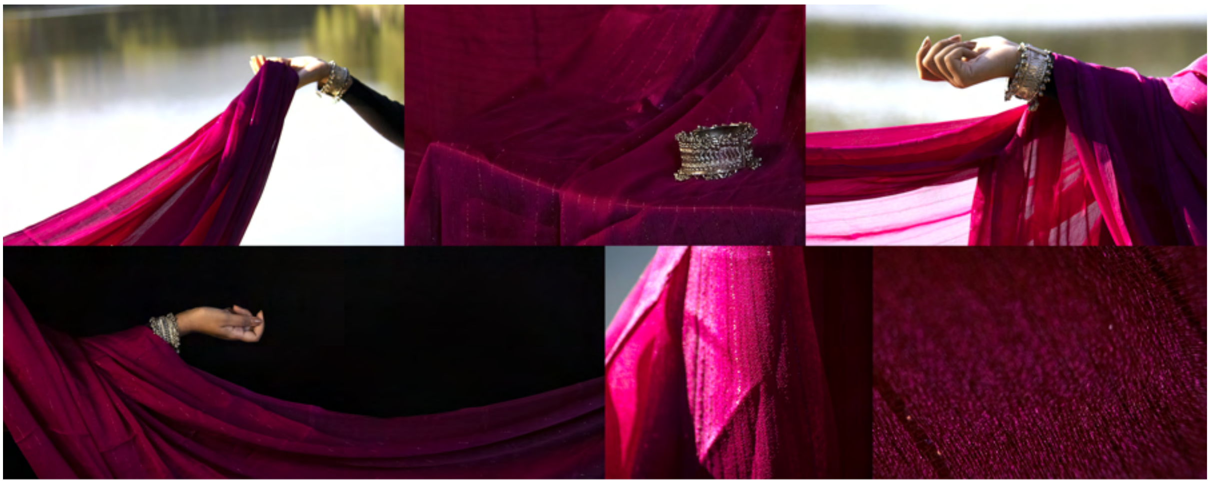

Image 8: Product photographs, clicked for user interviews.

In a second round of interviews, additional photographs sourced from the internet were used. These images featured unique compositions specifically intended to highlight different fabric properties. Participants were asked to interpret these images, providing further insights into which photographic techniques most effectively communicated specific material attributes.

Image 9: Product photographs for second round, shown to users.

The key findings are as follows

- Users often confuse fabric weight, color, and quality when viewing images alone.
- Shiny or transparent fabrics were frequently mistaken for party wear or cheaper materials due to lighting and drape.
- Inconsistencies in lighting led to incorrect assumptions about color, texture, and fabric feel.
- Close-up shots helped users understand texture and weave patterns more accurately.
- Multiple lighting conditions in different photos helped users form better judgments.
- Lack of human presence or scale references made it difficult to understand the length, width, or drape of the product.
- Products were sometimes mistaken for dupattas or scarves instead of sarees due to framing.
- Creases, folds, and fabric structure in the images contributed to perceived stiffness or smoothness.
- Backgrounds and surfaces influenced the perception of quality—messy or informal settings made products appear lower-end.
- Users judged comfort and usability (e.g., "itchy," "soft," "needs ironing") based on shine, texture visibility, and airiness in the photo.
- Several users mentioned they needed to see the backside, pallu, or edges to fully understand the saree.
- Decorative details like borders, motifs, and stitching were not always visible or emphasized.
- Users wanted to know what the fabric pairs with (e.g., jewelry, blouses, events) but couldn't judge from the photos.

Key insights of the research:

- **Raw Fabric Presentation:** Displaying fabric in its raw, unironed state, with natural creases, helps convey the material's texture and properties, offering a more authentic representation.

- **Highlighting Fabric Edges:** Including images that showcase the edges of the fabric is crucial, as it provides insights into the material's thickness, weight, and overall composition.

- **Weight and Flow Representation:** Demonstrating folds, bumps, and the fabric's natural flow can indicate its weight and structure, but many users still find it challenging to interpret these details accurately.

- **Close-Up Detail is Essential:** Close-up shots that reveal weave, texture, and stitch details are critical for helping users assess quality, craftsmanship, and comfort.

- **Variability Enhances Understanding:** Showing the same product under different lighting conditions and angles improves users' ability to judge fabric properties like sheen, texture, and fall.

- **Storytelling and Connection:** Sharing the story behind the fabric, its craftsmanship, and the artisans involved fosters an emotional connection with the product, increasing its perceived value and consumer engagement.

- Difficult properties to understand: Colour, Material, Quality and Flimsiness.

- Relatively easy properties to understand: Texture, Thickness and Transparency.

6. Phase 2: Define

6.1 Identified gap

In this phase, the insights gathered from desk research, social media analysis, and user interviews were synthesized to define the core problem and establish a clear direction for the design intervention.

The research revealed that while handloom products carry significant cultural and material value, artisans often struggle to represent these qualities effectively through digital media.

Through the synthesis of research findings and image analysis, three core components of effective product photography were identified: **alignment**, **lighting**, and **background**. These elements form the foundation of how a product is visually perceived in a digital context. However, several recurring issues were observed in how handloom artisans approach each of these aspects:

1. Alignment

- Fabrics are often placed haphazardly, leading to skewed or tilted frames.
- Edges and borders are not aligned properly, making the fabric appear unstructured or carelessly displayed.
- Important elements like the pallu or motifs are cut off or shown partially, failing to capture their full detail.

2. Lighting

- Photographs are often taken under poor lighting conditions, including dim indoor light or harsh midday sun.
- Direction of light is frequently ignored, resulting in uneven shadows or glare.
- Fabrics appear dull, overly shiny, or misleading in tone due to inconsistent lighting setups.

3. Background

- Cluttered or distracting backgrounds (e.g., household items, patterned floors) reduce the professionalism of the photo.

- Inconsistent or overly colorful backgrounds affect how fabric colors and textures are perceived.
- Absence of a neutral, consistent backdrop prevents the fabric from standing out clearly.

Identifying these recurring challenges helped define the direction for the design solution, which would focus on simplifying and guiding these three fundamental elements for artisans with minimal resources or training.

6.2 Design Brief

Designing a simple and accessible photography support system to help Indian handloom artisans effectively showcase their products in digital spaces with focus on improving how key fabric attributes are captured using basic smartphone setups.

6.3 Guidelines for Product Photography.

Based on the research findings, a set of **product photography guidelines** was developed to support handloom artisans in presenting their products more effectively through digital media. These guidelines were directly informed by the observed gaps in existing artisan photography practices, user feedback on digital fabric perception, and visual strategies used by established fashion brands.

1. **Use Close-Up Shots** – Capture fine details, textures, and craftsmanship of the fabric.
2. **Highlight fabric edges** - Capture the edge of the fabric bu covering 2/3 of the image with fabric.
3. **Shoot from Various Angles** – Showcase the product from multiple perspectives to highlight its design and structure.
4. **Experiment with Diverse Lighting** – Utilize natural and artificial lighting, Indoor and outdoor lighting, Front and side lighting, to bring out the true colors and depth of the fabric.
5. **Choose Complementary Backgrounds** – Use backgrounds that enhance the fabric's beauty without overpowering it.
6. **Embrace Natural Creases** – Photograph the fabric in its raw, unironed state to give a clear idea of material.

7. **Represent Weight and Flow** – Capture how the fabric drapes, folds, and flows to convey its weight and feel.
8. **Tell a Story** – Frame shots in a way that connects the fabric to its cultural heritage, artisans, and intended use.

7. Phase 3: Develop

7.1 Ideation

The ideation phase focused on generating potential solutions that could bridge the gap between how handloom products are currently presented and how they need to be visually communicated in digital spaces.

Concept- 1: VR Setup to Simulate Fabric in Virtual Environments

This concept proposed a fully immersive virtual reality (VR) system where users could experience handloom fabrics in a simulated 3D environment. The setup would allow for real-time fabric simulation, enabling consumers to observe how the fabric drapes, reflects light, and responds to movement. The goal was to replicate tactile understanding through digital immersion.

VR equipment and content creation require significant resources and technical expertise, which are generally not available to most handloom artisans in rural or semi-urban settings. Although impactful in enhancing digital fabric perception, this direction was not feasible for implementation in the context of artisans using basic smartphones.

Concept- 2: 3D Modeled Fabric Templates with Uploadable Designs

This concept involved the creation of pre-built 3D fabric templates that replicate realistic drape, fall, and texture. Artisans could upload flat images or scans of their fabric designs onto these templates to generate realistic visual mockups. These templates would simulate various lighting conditions, angles, and poses to give potential buyers a comprehensive sense of the product. Users can interact (rotation, zooming etc.) with these templates.

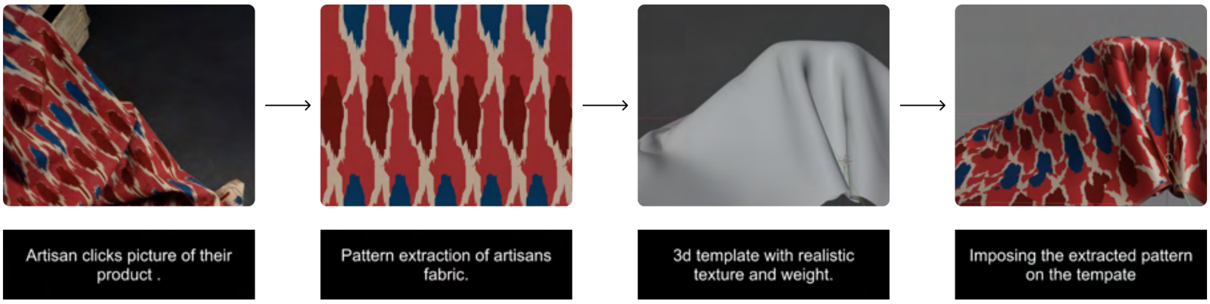

Image 10: Flow for concept 2.

However, it still requires artisans to digitize their designs and interact with 3D software, which may be beyond the skill level and resources of many in the target group. While the visual output could be strong, the process lacked the simplicity and smartphone-first accessibility that the brief demands. Handloom textiles are rich in tactile qualities, nuanced weaves, and material authenticity aspects that consumers often want to experience in their most real, unfiltered form. Unlike digitally native products, handloom items carry a cultural and material depth that is better communicated through authentic visual documentation rather than digital simulations or representations.

Concept- 3: Camera-Based Photography Guide with On-Screen Prompts and Tutorials.

This concept emerged as the most feasible and user-aligned direction. It proposes a **smartphone-integrated photography guide** that uses real-time **on-screen visual prompts** to guide artisans while capturing product images. The guide would assist with:

- Framing and alignment cues
- Lighting suggestions based on time of day and natural light
- Step-by-step composition support for showing fabric texture, borders, and drape

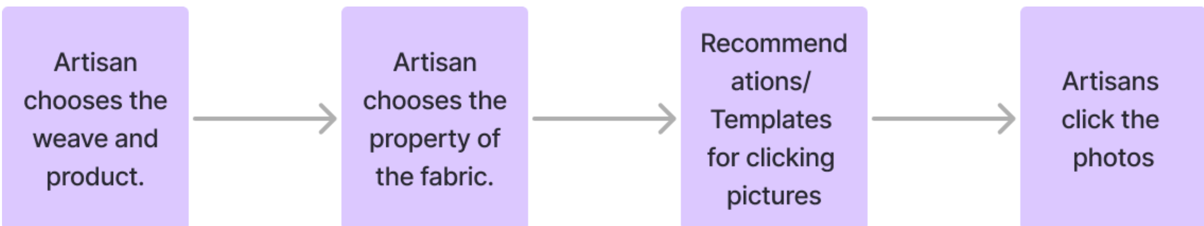

Image 11: Flow for concept 3.

Concept 3, the camera-based photography guide with on-screen prompts and tutorials, was chosen because it best aligns with both the needs of the handloom artisans and the expectations of consumers. It offers a simple, low-tech solution that fits seamlessly into the artisans' existing workflows using tools they already own, such as basic smartphones. Unlike simulation-based concepts, this approach preserves the authenticity of handloom products by capturing their real textures, colors, and drapes in natural settings.

7.2 How to use developed photography guidelines?

In order to develop effective photography templates, it was essential to first understand how each key fabric property could be visually captured through a smartphone camera, using the guidelines established during the define phase.

1. Colour
   - Use natural daylight or diffused artificial light to capture the true color. G4
   - Use a neutral or contrasting background to make the color pop. G5
   - Take multiple shots from different angles to see how the color appears under various lighting conditions. G3

2. Material
   - Capture close-up shots to highlight fabric weave or fiber structure. G1
   - Show different folds or drapes to emphasize material behavior. G7
   - Show natural creases of the material G6
   - Use side lighting to create soft shadows that define the fabric's texture. G4
   - Use props along with the fabric such as natural dyes, material yarn etc. G8

3. Quality
   - Take a macro shot of stitching, embroidery, or fabric weave to showcase craftsmanship. G1
   - Show the edges of the fabric G2
   - Capture the product in flow (e.g., a saree being draped or a curtain hanging naturally). G7
   - Include an artisan's hand touching the fabric to suggest quality perception.G8

4. Flimsiness
   - Hold the fabric up and let it fall naturally to show its structure. G7
   - Use side lighting to show how light passes through folds. G3

5. Texture
   - Use extreme close-ups to show fine details of the fabric surface. G1
   - Side lighting works best to emphasize shadows and depth. G3
   - Have a hand gently scrunching the fabric to highlight its texture.

- Use a contrasting smooth background to make the texture more visible. G5

6. Thickness
   - Fold multiple layers together and capture the thickness from the side. G7
   - Shoot from an edge perspective to highlight stacked fabric depth. G2
   - Place an object behind the fabric to demonstrate visibility through it. G5, G7

7. Transparency
   1. Hold the fabric against a light source to show how much light passes through. G4
   2. Place an object behind the fabric to demonstrate visibility through it. G5, G7
   3. Capture a shot with layered fabric to show varying transparency levels. G7

8. Pattern
   - Take flat-lay shots to display the full pattern without distortion. G3
   - Use close-ups to showcase intricate designs or weaving details. G1
   - Take a shot of the fabric draped or worn to show pattern placement. G7

9. Sheen/Gloss
   - Use angled lighting to highlight the shine or matte finish. G4
   - Capture a shot from multiple angles to show reflective properties. G3
   - Use dark backgrounds to enhance the contrast of the sheen. G5

10. Embroidery
    - Use close-up shots to capture embroidery threads, beads, or embellishments. G1
    - Side lighting helps in casting shadows that highlight embroidery depth. G4
    - Capture hands working on embroidery to add authenticity and craftsmanship. G8

7.3 Design

To streamline the photography process and make it more efficient for artisans, certain fabric properties that shared overlapping visual guidelines were combined. This allowed multiple attributes to be captured within a single composition, reducing the need for numerous separate shots. Based on this approach, a non-exhaustive list of templates was developed, with each template designed to highlight a specific set of fabric characteristics

Preset 1: Full Saree Display (Color, Pattern, Material)

- Grid Type: Rule of Thirds (3x3 grid)
- Product Placement: Saree should be spread out flat or draped over a surface.
- Purpose: Show the full saree to highlight color, pattern, and material.
- Guidelines:
  - Ensure the saree covers most of the frame.
  - Keep the top border aligned with the top third of the grid for a balanced composition.
  - If draped, the pleats should align with vertical grid lines.

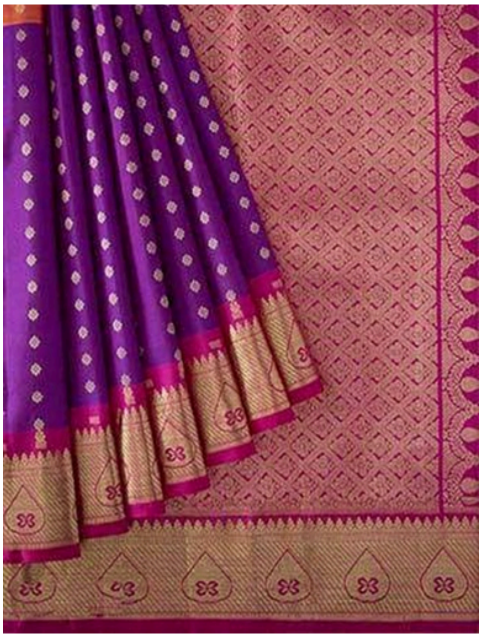


Image 12: Preset 1(left) and Preset 2(Right) reference image

Preset 2: Close-up of Texture & Weave (Texture, Thickness, Material, Transparency)

- Grid Type: Center Focus Grid (Guides artisans to focus on a small section)
- Product Placement: A well-lit section of the saree, preferably in natural light.
- Purpose: Capture texture, thickness, and weave quality.
- Guidelines:
  - Ensure the saree fills most of the frame.
  - Keep texture details in the center grid.
  - Use soft light to avoid harsh reflections.

Preset 3: Draped Look (Flimsiness, Sheen, Flow, weight)

- Grid Type: Leading Lines Grid (Diagonal guidelines to guide fabric flow)
- Product Placement: Drape the saree over a hanger, bamboo, or mannequin.
- Purpose: Showcase how the saree flows, its flimsiness, and weight.
- Guidelines:
    - Let the saree fall naturally.
    - Position folds or drapes along the diagonal grid lines.
    - Use a side light source to capture sheen.

Image 13: Preset 3 (left) and Preset 4 (Right) reference image

Preset 4: Embroidery & Border Details (Embroidery, Quality)

- Grid Type: Detail Frame Grid (Small rectangle for embroidery focus) and leading lines
- Product Placement: Close-up of the saree border or an embroidered section.
- Purpose: Highlight embroidery details and fabric quality.
- Guidelines:
    - Keep embroidery inside the highlighted frame.
    - Use side lighting and ensure the details are sharp and well-lit.
    - Use a contrast background for better visibility.

Preset 5: Folded Stack (Thickness & Material)

- Grid Type: Horizontal and diagonal Lines Grid (For aligning folds)
- Product Placement: Stack the saree neatly with visible folds.
- Purpose: Show thickness and material weight.
- Guidelines:
    - Keep folds parallel to horizontal grid lines.
    - Use side lighting for depth.
    - Make sure the edge of the saree is visible for thickness reference.

Image 14: Preset 5 reference image

7.4 Flow of the process

The platform was designed to guide artisans through a simple, step-by-step process for capturing high-quality product photographs.

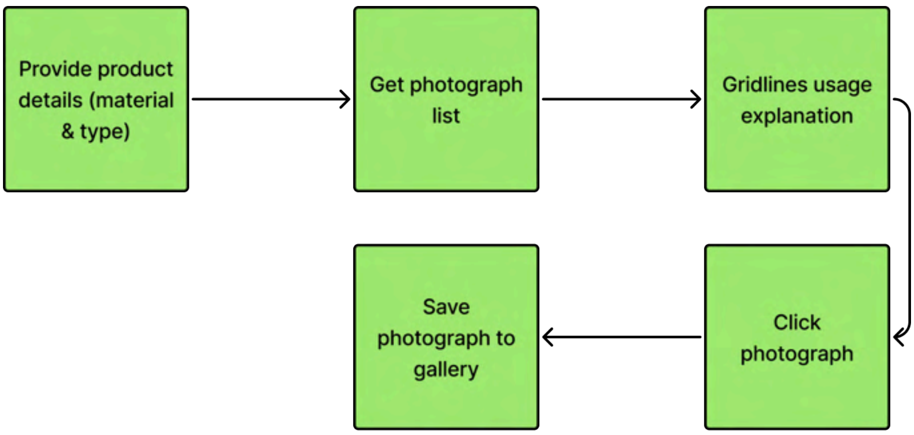

Image 15: User flow to click photograph

7.5 Screens

High-fidelity mobile screens were designed and prototyped to represent the chosen concept since mobile is most accessible for artisans. The user interface was intentionally kept simple and intuitive, with the use of large, bold text to ensure readability and easy comprehension for artisans. Button sizes were increased to enhance tap accuracy and usability, especially for users who may not be familiar with smartphone navigation.

Additionally, the app was designed to be presented in the local language of the artisan.

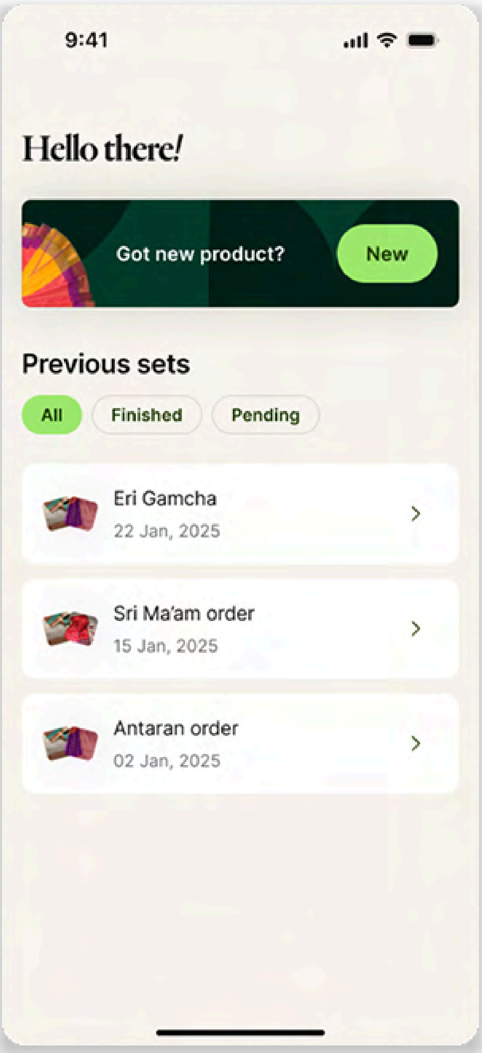

Image 16: Home Screen

When the artisan opens the app, this home screen will be visible. They can quickly start photographing a new product by tapping the "New" button at the top. Below, they can view all their previous photo sets, filtered by status, All, Finished, or

Pending. Each entry shows the product name, date, and a thumbnail, making it easy to track and manage their work.

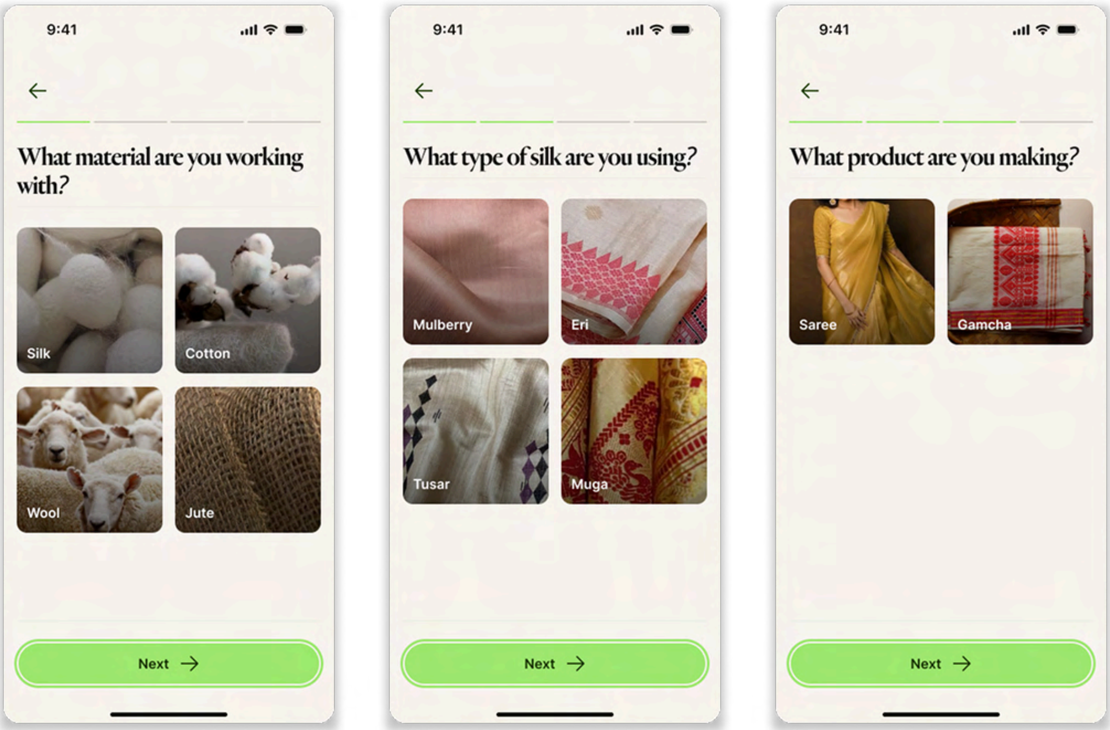

Image 17: Product selection screens

When the artisan starts a new product shoot, this flow helps them choose the type of product they are working with. They begin by selecting the material (such as silk, cotton, wool, or jute), then the specific type (for example, eri or muga silk), and finally the product category (like saree or gamcha). This step-by-step selection filters out the most relevant photography templates, ensuring the visual guidance is tailored to the specific fabric and product.

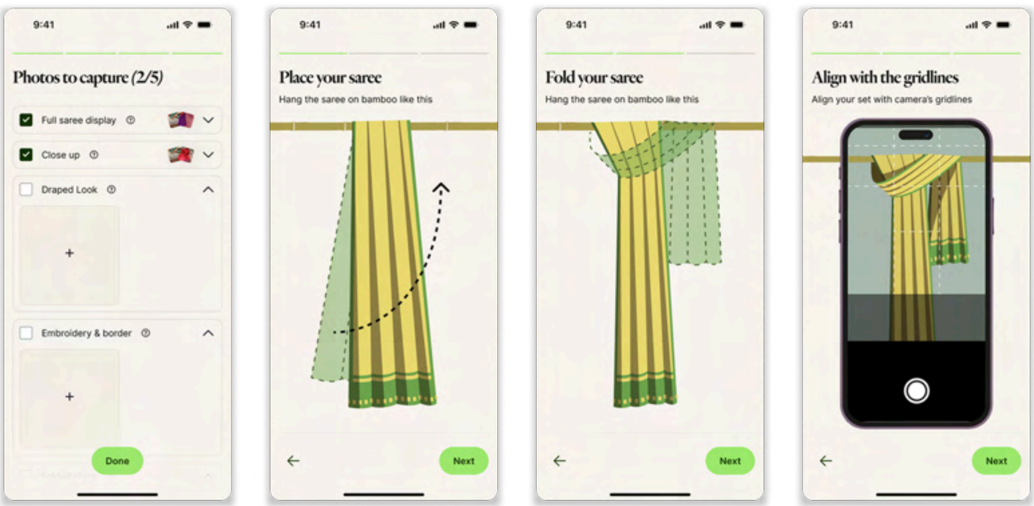

Image 18: Template guide

Once the artisan selects a template, a sequence of illustrated guides appears to help them set up the fabric correctly. These step-by-step visuals show how to place, fold, and position the fabric.

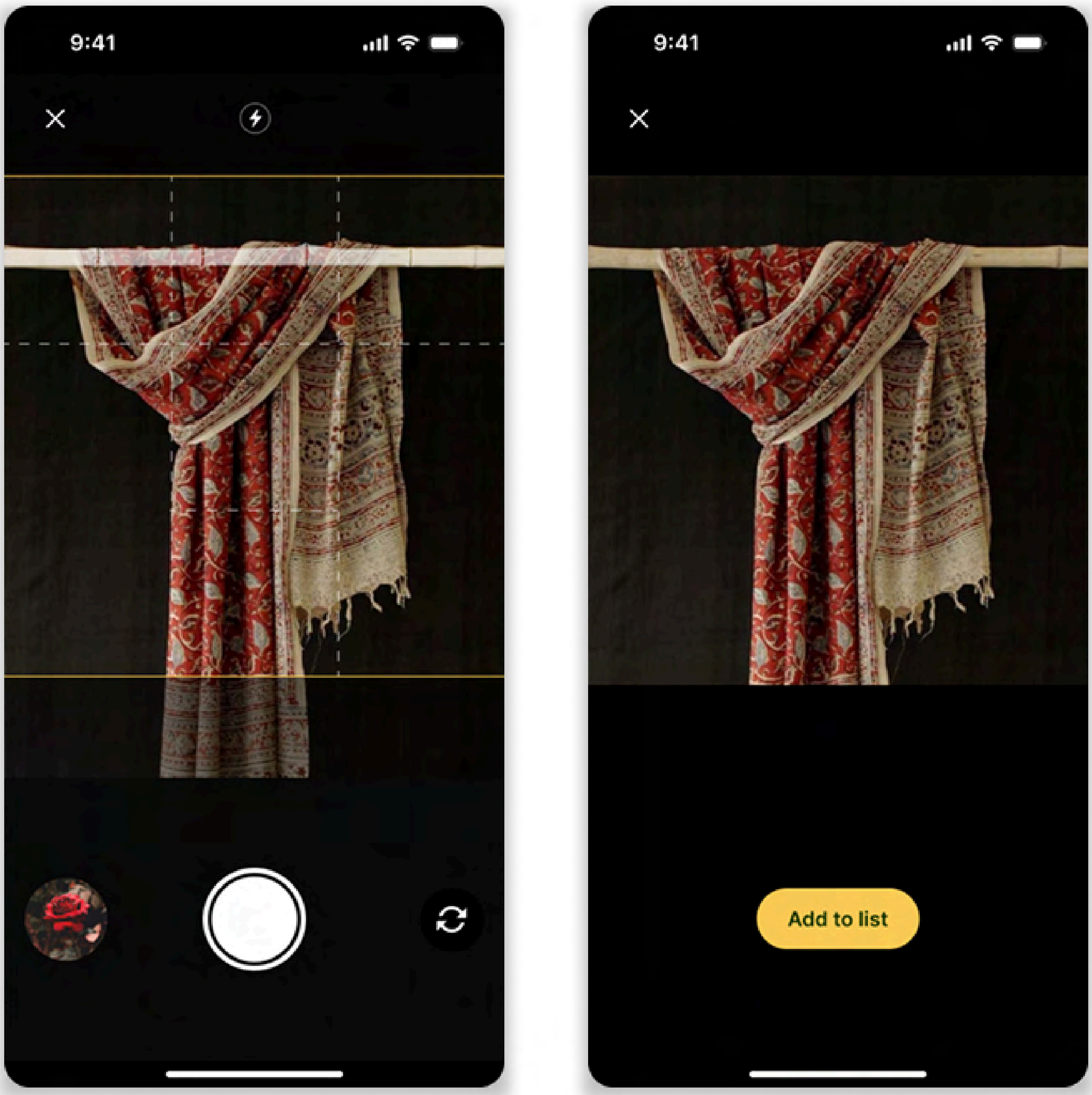

Image 19: Camera screens

Finally, a live camera view with on-screen gridlines helps them align the fabric accurately before clicking the photo. This visual guidance ensures the fabric is captured clearly and consistently, highlighting its important properties as intended.

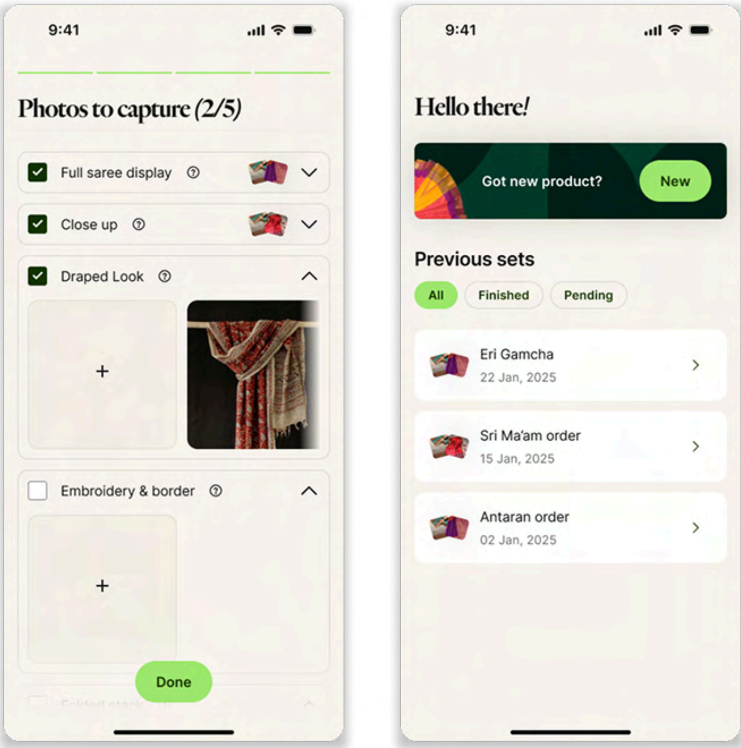

Image 20: Saved photos screen

After clicking each photo, the image is automatically saved both within the app and in the artisan's mobile gallery for easy access. A built-in to-do list tracks the required photo types and ensures that the artisan captures all necessary shots.

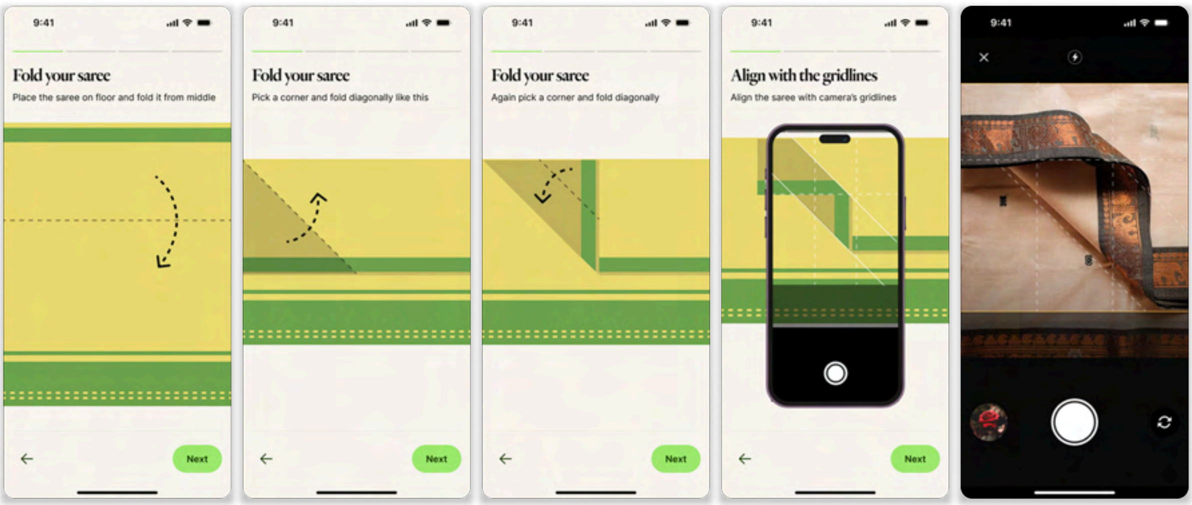

Image 21: Template 2 user flow

This screen sequence illustrates the user flow for a folding-based template. Once the artisan selects this template, the app provides a series of clear, illustrated instructions guiding them through the correct method of folding the fabric.

8. Phase 4: Deliver

8.1 User Testing

To evaluate the usability and effectiveness of the prototype, user testing sessions were conducted with eight female handloom artisans from the Kamrup cluster in Assam. The objective was to understand how well the platform supports artisans in photographing their products using the provided templates and visual guidance.

Demographic Overview:

- All participants were active handloom artisans.
- 5 out of 8 owned smartphones; 3 had never used a smartphone before.
- Only 1 participant had prior experience in photographing her products for digital use.

Localization Efforts:

To ensure clarity and accessibility, the prototype interface was fully translated into Assamese, the local language of the participants. This allowed users to engage with the content more comfortably and reduced reliance on interpretation.

Prototype Setup:

For testing the template layout feature, interactive filters were created using Snapchat's Lens Studio. These filters mimicked the final product by displaying on-screen visual guides, such as placement instructions and alignment frames.

Testing Process:

Participants were asked to follow the instructions displayed in the app and use the guided templates to click photographs of their own handloom products. They were observed as they navigated through the flow, selected templates, arranged their fabrics, and captured the required images.

Image 22: User testing conducted with handloom artisans.

Findings of user testing:

Artisans required additional verbal guidance to correctly set up the product or saree for photography. Most did not read the on-screen instructions and instead relied on the visual diagrams, which proved insufficient without detailed context, such as where to place the saree border or how to orient it. Even seemingly obvious steps, like pressing the edges to prevent them from curling, were not intuitively followed. However, once one artisan was guided through the process, the others quickly learned by observing, and were able to proceed without referring back to the instructions.

A key challenge was identifying good lighting. Despite clear daylight outside, artisans often attempted to take photos indoors in poor lighting conditions. They lacked an understanding of what constitutes "good light" and where to position the product.

In the second setup, which included a bamboo frame, artisans struggled to align the physical bamboo with the guideline on the screen. They did not realize the line was meant to assist with alignment. This was compounded by the fact that the guides were rendered at low opacity, making them hard to see in bright sunlight.

On a positive note, artisans were able to align the product correctly using the grid in the first setup, which did not include any external objects.

Image 23: Photographs clicked by artisans during user testing

Identified problems:

- Reliance on text-based instructions is ineffective.
- Even photos/diagrams alone didn't communicate detailed actions clearly enough.
- Key details (e.g. border placement, edge flattening) were not obvious from visual cues.
- Learning is social and observational instructions weren't self-explanatory or self-sufficient.
- Artisans struggled to identify good lighting conditions.

- No real-time feedback or cues were given on lighting quality.
- Low visual contrast/opacity made guides confusing and hard to distinguish.
- No clear cue that they were supposed to align with the guide.
- UI not tested for outdoor visibility sunlight drastically reduces visibility of soft overlays.
- Simple grid-based guidance without external props (like bamboo) worked best.
- Shows potential of minimal UI with visual alignment cues.

Possible solutions for these problems:

Highest Priority

- Replace text-heavy screens with video instructions or animated sequences showing step-by-step action (especially for folding/orienting).
- Add light quality detection with real-time guidance: e.g., "This light is too dark — try moving near a window or outside."

High Priority

- Introduce a demo mode: one guided video that explains the entire process for the first user.
- Add "pulse" or blink animation showing alignment zone.
- Consider a toggle for indoor/outdoor mode, adjusting overlay brightness automatically.

Medium Priority

- Add voiceover in Assamese to supplement visuals — even short and direct.
- Add peer-learning cues like: "Ask a fellow artisan if you're not sure!" (This validates learning by watching others.)
- Add visual samples of good vs. bad lighting during onboarding.
- Possibly break setup into micro-tasks: First grid → then bamboo → then lighting.

Low Priority

- Consider using "tap-to-play" audio guides that explain what's in the image.
- Make guides bolder or animate briefly when first visible.

- Use icons or sun/cloud graphics to show recommended light conditions (e.g., “Morning sunlight near window” vs. “Dark corner indoors”).
- Use highlighted contrast colors, not subtle outlines, especially in outdoor sunlight.
- Add arrow labels or short instructions next to the guide (e.g., “Align your bamboo here”).
- Redesign guides with high-contrast, thick lines for outdoor use.
- Offer short colored indicators (e.g., red line turns green when aligned).
- Keep interface minimal and clear.
- Progressively introduce more complexity (e.g., add bamboo after mastering plain alignment).

8.2 Changes in the solution

Based on the identified problems and feedback received during user testing, several refinements were made to the proposed solution to better suit the needs and capabilities of the artisans.

One significant change was replacing the illustrated tutorials with video-based tutorials. During testing, it was observed that users found it difficult to interpret and follow static illustrations accurately. To address this, the revised solution includes short, one-minute instructional videos for each template. These videos are translated into the artisan’s local language to ensure clarity and ease of understanding.Each video clearly demonstrates key steps, such as how to position the fabric, where and how much to fold, how to align borders, and techniques for flattening fabric to avoid bumps or creases. The aim is to provide clear, relatable, and repeatable instructions that can be easily followed by artisans with varying levels of literacy and digital familiarity.

Image 24: Snapshots from template-1 tutorial video

The second change in the solution was the inclusion of clear, visual instructions related to lighting conditions for capturing product photographs. Since lighting was identified as a critical factor affecting fabric visibility, color accuracy, and texture representation, it was necessary to guide artisans on how to achieve optimal lighting using natural sources.

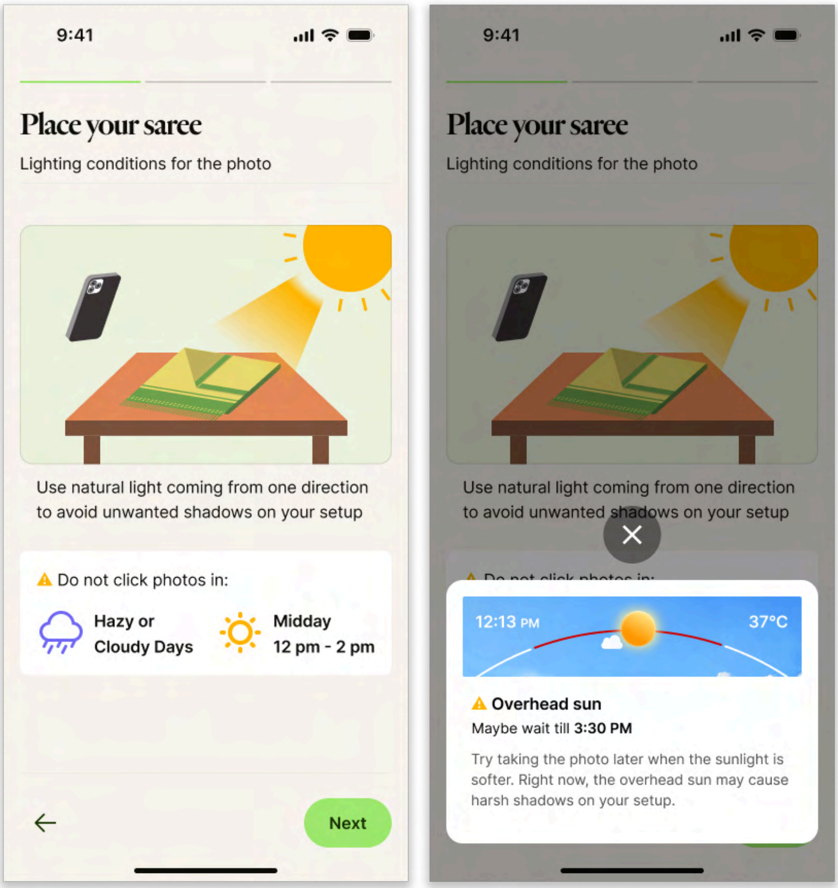

Image 25: Instructional screens for lighting conditions

An illustration was added to demonstrate the recommended setup, advising artisans to click photographs outdoors in natural sunlight, with the phone positioned in a way that avoids casting shadows onto the fabric. This visual cue makes it easier for users to understand the ideal placement and direction of light.

In addition to the illustration, the screen also includes a written advisory about weather and time-related considerations. Artisans are guided to avoid hazy or cloudy days, when light is too dim, and to not shoot between 12 PM and 3 PM, when sunlight is excessively harsh and can result in overexposed or washed-out images.

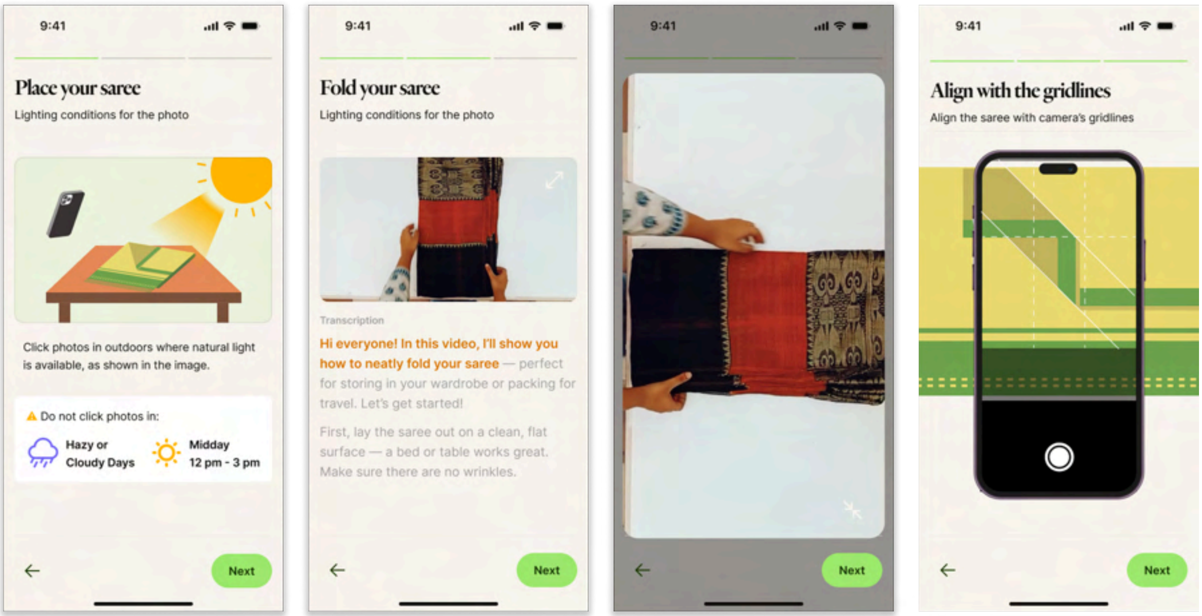

Image 26: Instruction sequence including video tutorial

Revised Instructional Sequence

To improve clarity and ease of use, a new three-step instructional sequence was introduced to guide artisans before they capture product photos. This sequence was designed based on user feedback.

- Screen 1: Lighting and Setup Instructions
  This screen as explained before provides guidance on the optimal lighting conditions for photography. An illustration shows the correct placement of the fabric and the phone in relation to the sun, helping users avoid shadows and glare. Additionally, a weather advisory is included.

- Screen 2: Video Tutorial with Transcript
  The next screen features a short instructional video demonstrating how to set up and photograph the fabric for the selected template. The video also includes a live transcript displayed on-screen for better comprehension. Users can also expand the video to full screen for a clearer view of each step.

- Screen 3: Alignment Illustration
  The final screen presents a clear illustration showing how the fabric should be aligned with the on-screen camera gridlines. This visual aid ensures proper framing and helps artisans position their product consistently and accurately for optimal results.

Another key improvement in the solution was the integration of live visual cues within the camera interface to assist artisans during the photo-capturing process. The camera now includes a system that detects whether the fabric is properly aligned with the on-screen gridlines. If misalignment is detected, real-time text prompts appear to guide the user in correcting the setup.

These prompts include cues such as:

- **“Move closer”** – when the product is too far away.
- **“Move further from subject”** – when the camera is too close, cutting off parts of the fabric.
- **“Center your subject”** – when the fabric is off-center.
- **“Backlight detected”** – when strong light from behind the fabric could reduce clarity.
- **“Ready”** – when alignment and lighting are optimal for capture.

This feature ensures that even users with limited photography experience receive instant, actionable feedback, leading to better-composed and visually accurate product photos.

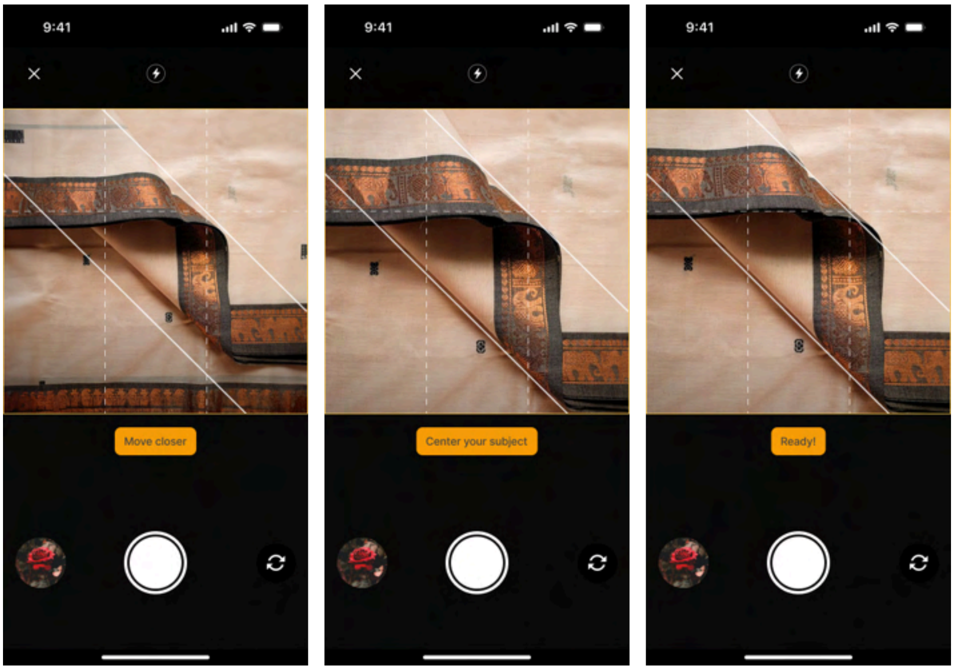

Image 27: Live prompts on camera

9. Conclusion

The project successfully addresses a key barrier preventing handloom artisans from fully participating in the digital economy: the inability to present their products in a visually compelling and informative way. By understanding user perception, analyzing the practices of successful fashion brands, and identifying common mistakes in artisan photography, the project defines clear opportunity areas for intervention. The final solution, an intuitive, template-based photography guide with real-time camera support and localized instructional videos, proves to be a practical and scalable approach. User testing with artisans in Assam confirmed that the redesigned flow and features were more accessible, easier to follow, and led to better-quality images. This system not only improves the digital presentation of handloom products but also empowers artisans to take control of how their craft is represented and perceived online.

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